Creating an axis cross

Creating an Axis Cross in Matplotlib

An axis cross means arranging the x-axis and y-axis so that they intersect at the origin ((0,0)) instead of having the x-axis at the bottom and the y-axis at the left edge of the plot.

Normally, Matplotlib gives us something like:The x-axis is at the bottom and the y-axis is at the left. But for mathematical functions, it is often preferable to have:the two axes cross at ((0,0)).

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-5, 5, 200)
y = x**2

fig, ax = plt.subplots()

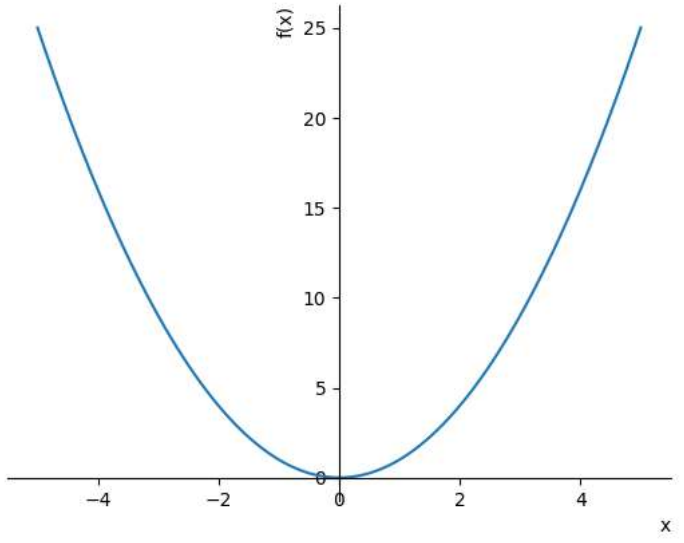
ax.plot(x, y)

# Move y-axis to x = 0
ax.spines['left'].set_position('zero')

# Move x-axis to y = 0
ax.spines['bottom'].set_position('zero')

# Hide top and right spines
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.set_xlabel('x',loc='right')
ax.set_ylabel('f(x)',loc='top')

plt.show()
```




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import matplotlib.pyplot as plt

x = np.linspace(-5, 5, 200)
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fig, ax = plt.subplots()

ax.plot(x, y)

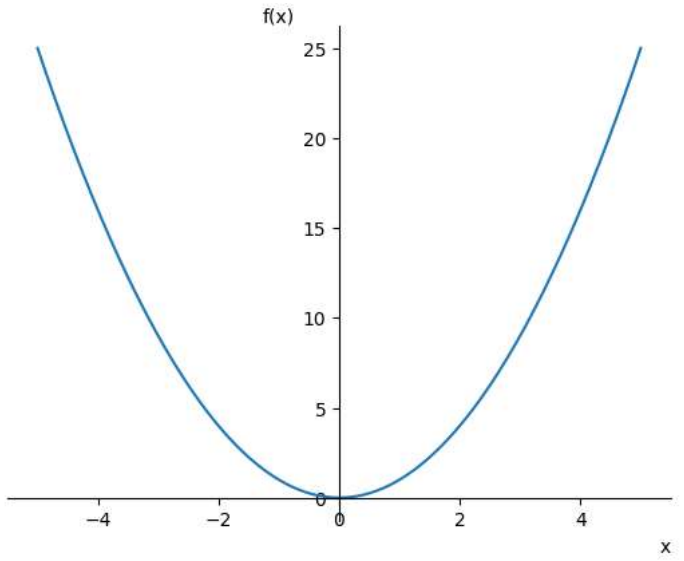
# Move y-axis to x = 0
ax.spines['left'].set_position('zero')

# Move x-axis to y = 0
ax.spines['bottom'].set_position('zero')

# Hide top and right spines
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
```

```
ax.set_xlabel('x',loc='right')
ax.set_ylabel('f(x)',loc='top',rotation=0)

plt.show()
```



changing the font size and color of the label text

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-5, 5, 200)
y = x**2

fig, ax = plt.subplots()

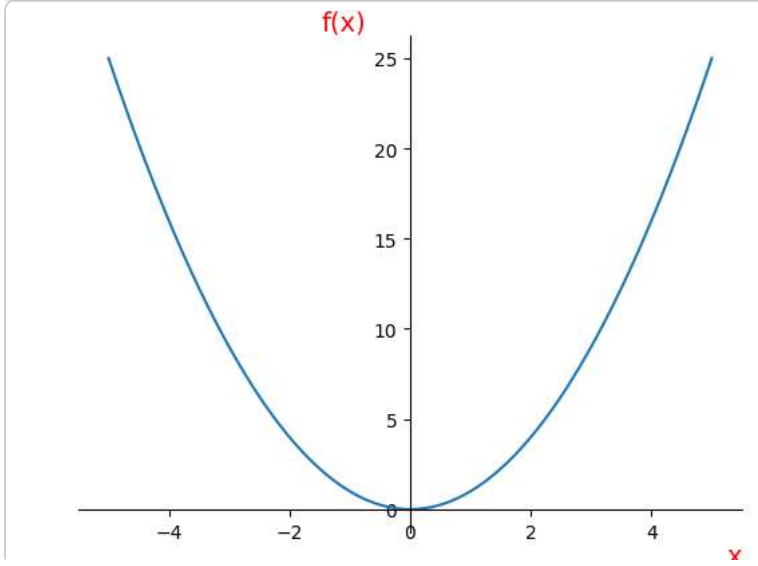
ax.plot(x, y)

# Move y-axis to x = 0
ax.spines['left'].set_position('zero')

# Move x-axis to y = 0
ax.spines['bottom'].set_position('zero')

# Hide top and right spines
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.set_xlabel('x',fontsize=14,color='r',loc='right',labelpad=1)
 # set fontsize and colour of the label
ax.set_ylabel('f(x)',fontsize=14,color='r',loc='top',rotation=0)

plt.show()
```


f(x)
25
20
15
10
5
0
−4
−2
0
2
4
x